Noninvasive and Quantitative Measurement of Intrinsic Laryngeal Muscle Activity Using Surface Electromyography

Xinyi Yang, Jing Yan, Hanting Zhang, and Liang Wu

***Abstract*— *Objective:* Healthy functions of intrinsic laryngeal muscles (ILMs) are important for normal speech production. This work aims to measure ILM activities for accurate assessment of laryngeal function. *Methods:* A noninvasive method using two-channel surface electromyography (sEMG) was designed to extract activity signals of cricothyroid (CT) and lateral-cricoarytenoid (LCA) muscles. The extracted sEMG signals were verified by the clinical laryngeal electromyography (LEMG) signals and quantified by time-domain and frequency-domain features. The sEMG features were further compared with acoustic features (i.e., fundamental frequency *F*0 and closed quotient (CQ) of speech signals to investigate dynamic changes in muscle activity during speech control. *Results:* The extracted sEMG signals captured consistent features of muscle activity with the LEMG signals. The frequency-domain features of CT/LCA muscles had a significant positive correlation with *F*0 but not with CQ, indicating an increasing activation of CT/LCA muscles with voice pitch rising. Furthermore, CT muscle showed greater changes in the frequency-domain features than LCA muscle during pitch control. These outcomes were consistent with the effect of CT/LCA muscles on voice control reported in previous studies. Moreover, the frequency-domain features had a more stable pattern than the time-domain features across subjects, showing potential as reliable parameters for quantitative assessment of laryngeal muscle activity. *Conclusion:* The feasibility and reliability of the proposed method in the extraction and evaluation of dynamic muscle activity during phonation were verified. *Significance:* The proposed method is promising to develop into a practical technique for in vivo human research on neuromuscular control and clinical diagnosis of laryngeal disorders.**

***Index Terms*— Frequency-domain feature, intrinsic laryngeal muscle (ILM), muscle activity, speech control, surface electromyography (sEMG).**

I. Introduction

SPEECH production is highly dependent on the activities of intrinsic laryngeal muscles (ILMs), which determine speech features (e.g., pitch and voice type) through modulating laryngeal posture and vocal fold mechanical properties [1], [2]. Healthy function and coordination of ILMs are essential for normal speech production, while laryngeal muscle dysfunction leads to vocal fold immobility, voice abnormality, and even more severe cases like dyspnea [3], [4]. Therefore, it is necessary to investigate the effect of ILM activation and coordination on phonation, which not only contributes to a better understanding of neuromuscular control of voice production and quality but also helps in the diagnosis of laryngeal neuromuscular disorders. However, proper measurement and quantitative evaluation of laryngeal muscle activities are prerequisites for all studies and applications.

Due to the technical challenges of human in vivo experiments, most previous studies were based on ex vivo or in vivo animal models, in which external electrical stimulation was applied to laryngeal nerves to produce muscle activity [5], [6]. These models allow for the investigation of the isolated functions of individual laryngeal muscles, as well as the interactions of multiple muscles in controlling glottal posture and voice quality [7], [8]. However, passive muscle activation cannot provide insight into the neuromuscular mechanism of human phonation during various vocal tasks. This aspect must be studied in a human in vivo model during actual speech production.

Laryngeal electromyography (LEMG), because of its direct detection of ILM activity in response to laryngeal nerve control by inserting needle electrodes into individual laryngeal muscles [9], is currently the most commonly used method in human measurement, promoting research on the regulation role of laryngeal muscles in pitch, intensity, and register during normal phonation [10], [11], [12]. Moreover, LEMG can also provide a basis for the identification of muscle fatigue, vocal fold paralysis, and nerve injury [13], [14], [15], thus making it a standard tool in the clinical diagnosis and prognosis of laryngeal neuromuscular diseases [16]. However, as an invasive technology, the insertion of needle electrodes causes a temporary injury to the laryngeal muscle and considerable discomfort to the patient [17]. This can impact muscular control during normal speech production, finally affecting the accuracy and reliability of the measurement. In addition, due to the small size and complex structure of the larynx, performing LEMG is technically difficult, which requires the operator to have high skill for needle placement and good experience in neurophysiology [18].

Compared to needle EMG, surface electromyography (sEMG) is a noninvasive, highly acceptable, and more practical

Received 28 January 2025; revised 30 March 2025; accepted 19 April 2025. Date of publication 5 May 2025; date of current version 15 May 2025. This work was supported by the National Natural Science Foundation of China under Grant 12274340. The Associate Editor coordinating the review process was Dr. Ting-Wei Wang. *(Corresponding author: Liang Wu.)*

Xinyi Yang, Hanting Zhang, and Liang Wu are with the Key Laboratory of Biomedical Information Engineering of Ministry of Education, Department of Biomedical Engineering, School of Life Science and Technology, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China (e-mail: liangwu@xjtu.edu.cn).

Jing Yan is with the Department of Otorhinolaryngology Head and Neck Surgery, The Second Affiliated Hospital, Xi'an Jiaotong University, Xi'an 710049, China.

Digital Object Identifier 10.1109/TIM.2025.3566810

1557-9662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

method to detect electrophysiological activity of the whole muscle or muscle group [19]. It has also been used to measure muscle activity in the neck area for vocalization evaluation [20], voice disorder diagnosis [21], [22], and speech recognition [23], [24]. Regarding the laryngeal muscle activity, for example, Bracken et al. used 2-D spectral energy maps of sEMG signals to identify differences in anterior neck muscle activity between different phonation conditions [25], while Stepp et al. [4] reported low coherence in the sEMG signals of laryngeal muscle activity in vocal hyperfunction patients. It has been shown that sEMG has potential in the evaluation of laryngeal muscle activity during phonation, but these studies primarily focused on the analysis of sEMG signals from the whole laryngeal muscle group or superficial external laryngeal muscles like the sternohyoid (SH), sternothyroid (ST), and thyrohyoid (TH) muscles. Thus, no study has as yet been conducted to extract specific muscle activity from the sEMG signals and investigate the contraction pattern of each ILM during voice control.

Although recorded sEMG signals are generally a combination of electrical signals from several muscles, different muscles generate different spatial and temporal distributions of sEMG signals due to different electrical signal generation locations and anisotropic propagation [26], [27]. Thus, it is possible to separate a single ILM signal from the combined sEMG signals by taking advantage of differences in muscle position and fiber orientation. In this work, a two-channel sEMG acquisition and processing method was designed to measure the electrical activities of two specific ILMs, i.e., cricothyroid (CT) and lateral-cricoarytenoid (LCA) muscles. The CT and LCA muscles were selected because they play important roles in voice control, are frequently tested in the diagnosis of laryngeal diseases, and are located at a slightly superficial position to produce a detectable signal [28], [29], [30]. The contraction of CT and LCA muscles was quantified using time-domain and frequency-domain features based on the extracted sEMG signals of each muscle. The goal was to verify the validity of the method for measuring dynamic CT and LCA muscle activity during phonation. The details of the method and validation are described in Sections II and III. Then, CT and LCA muscle activities are quantitatively evaluated and compared with acoustic features in Section IV, and the results are discussed regarding the feasibility and accuracy of the method and features for CT and LCA muscle activity measurement in Section V.

II. Methods

A. Data Collection

1) Subjects: Data were collected from nine male and seven female subjects, ranging in age from 21 to 39 (averaged 24.69 ± 4.03) years old. All subjects reported no history of laryngological or speech disorders. Each recruited subject received an explanation of the experimental purpose and procedures before the study, and informed consent was obtained verbally before the publication of any data included in this article. The experimental protocol was in accordance with the declaration of Helsinki and approved by the Ethics Committee of Xi'an Jiaotong University Health Science Center (no. 2022–1080).

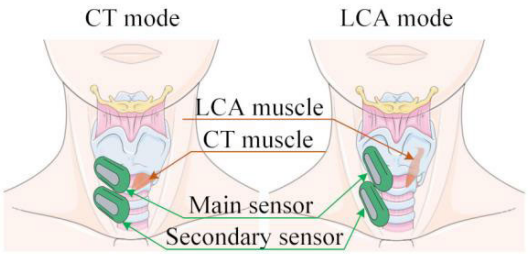

Fig. 1. Two modes of sensor placement. (The figure was partly generated using Servier Medical Art, provided by Servier, licensed under a creative commons attribution 3.0 unported license.)

2) Data Acquisition: Two channels of sEMG signals were simultaneously captured by two differential sensors using a programmable signal acquisition device (DataLINK, Biometrics Ltd., U.K.). The technical specifications of the differential sEMG sensor were detailed in Table I. To ensure reliable and accurate signal acquisition, the DataLINK system underwent comprehensive calibrations, including gain calibration, frequency response calibration, zero calibration, and noise calibration.

One main sensor was used to collect sEMG signals from the target muscle, and the secondary sensor was used to record sEMG signals from surrounding interfering muscles. Each sensor was constructed of two circular electrodes with a diameter of 10 mm and a center distance of 20 mm. Before placing each sensor, the neck skin of the subject was cleaned with water to remove any sweat and oil, and then the sensor was attached to the neck skin using the die-cut medical-grade double-sided adhesive.

As target muscles in this study, CT and LCA muscles were anatomically close to each other. In order to maximize the differences in sEMG signals between the two muscles, two modes of sensor placement (i.e., CT mode and LCA mode) were designed according to the different orientations of the CT and LCA muscles, as shown in Fig. 1. For each mode, the main sensor was placed above and along the CT or LCA muscle, making sure that the line between the centers of the two electrodes of the sensor was parallel to the muscle orientation. The secondary sensor was placed close to the lower edge of the main sensor with the same orientation to record interfering signals from the SH and ST muscles.

The orientation of each muscle was determined by palpation. For CT muscle, the cricoid cartilage was found first, then the lower edge of thyroid cartilage was above the cricoid cartilage, and the cornu inferius could be found backward along the lower edge. The axial direction of CT muscle was determined by the connecting line from the middle part of cricoid cartilage to the cornu inferius. For LCA muscle, the V-shaped incisura of thyroid cartilage was first palpated, and the tuberculum thyroideum superius was found posteriorly along the superior edge of the thyroid cartilage. The line from the midpoint of the tuberculum thyroideum superius and tuberculum thyroideum inferius to the lateral edge of the cricoid cartilage was considered to be the axial direction of LCA muscle fibers.

3) Experimental Settings: Due to the overlap in sensor placement between the two modes, it was impossible to obtain sEMG signals from CT and LCA muscles on the same side of the larynx. As a result, three sets of experiments were carried

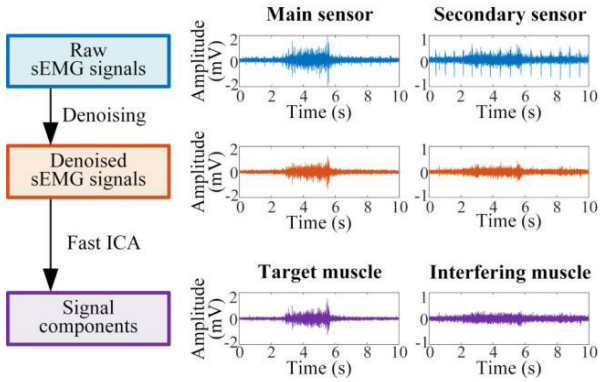

Fig. 2. sEMG signal processing.

out to investigate the dynamic activities of different muscles. In the first two sets of experiments (i.e., Exp 1 and Exp 2), CT and LCA muscles were separately and independently studied. For each experiment, sEMG signals from both sides of the larynx were collected with the same sensor placement mode to evaluate the similarity of muscle activities on both sides. In the third experiment (i.e., Exp 3), sEMG signals of CT and LCA muscles were synchronously collected from the right and left sides, respectively, to compare the differences in the activities of CT and LCA muscles.

All subjects participated in three sets of experiments. According to guidelines of the European Laryngological Society for clinical use of LEMG and research into laryngeal muscle control mechanisms, vowel/i:/was often recommended in clinics to evaluate vocal function and muscle coordination due to its ability in effective activation of CT and LCA muscles [16], [31]. Also, the tongue position and vocal tract shape during vowel/i:/production were relatively consistent across subjects, minimizing variability in muscle activity and sEMG data. Therefore, in all experiments, the subjects were asked to produce vowel/i:/in an increasing pitch manner within their comfortable vocal range, i.e., chest voice. Each phonation lasted 5–7 s and was repeated five times with a 30-s rest interval to avoid muscle fatigue. To reduce the influence of extrinsic laryngeal muscles, participants were instructed to maintain a steady head posture and perform stable respiratory support during the trial.

During phonation, sEMG signals were collected at a sampling frequency of 1000 Hz, while speech signals were synchronously recorded with a dynamic microphone mounted 20 cm away from the mouth. In total, 240 sets of data (16 subjects × 3 experiments × 5 phonations) were obtained for processing and analysis in this study.

B. sEMG Signal Processing

Two processing steps were applied to extract the sEMG signal of the target muscle from the two-channel raw sEMG signals. As shown in Fig. 2, the two processing steps, including signal denoising and extraction, are illustrated with a typical example.

1) sEMG Signal Denoising: First, raw sEMG data from each channel was preprocessed through a Butterworth bandpass filter with cutoff frequencies of 30–450 Hz. Second, to remove the electrocardiograph (ECG) noise, a method combining wavelet transform (WT) and fast independent component analysis (Fast ICA) was used to extract the ECG signals from the filtered sEMG signals [32]. The WT decomposed each channel of the sEMG signal into a number of subsignals in different frequency bands using the wavelet basis function of Symlet4 and the decomposition scale of 8. The Symlet4 had a good symmetry to minimize phase distortion during signal analysis and reconstruction, while the 8 decomposition levels ensured sufficient channel data and avoided loss of signal details caused by excessive decomposition. Then, ECG signals were estimated from the subsignals by Fast ICA. As a result, ECG noise was effectively removed from the filtered sEMG signals, especially the signal from the secondary sensor, as shown in Fig. 2.

2) Extraction of Target Muscle sEMG: Although the main sensor was placed above the target muscle, the sEMG signals of the target muscle were interfered with the electrical activity signals of the surrounding muscles, which were collected by the secondary sensor. Fig. 2 shows obvious differences in waveforms from the two sensors, indicating electrical signals of different muscle activities. Therefore, Fast ICA was used to extract sEMG signals of the target muscle from the two channels of denoised sEMG signals.

First, two independent signal components (denoted as C_1 and C_2) were decomposed using Fast ICA from the two channels of sEMG signals, as shown in Fig. 2. Since the main sensor was placed close to the target muscle, it was believed that the main sensor captured more electrical signals from the target muscle than the secondary sensor did. Then, the relationships between the two signal components (C_1 and C_2) and the sEMG signal from the main sensor (S_{main}) were quantified by Pearson's correlation coefficients $r(C_1, S_{\text{main}})$ and $r(C_2, S_{\text{main}})$. Finally, the signal component with a higher correlation coefficient was identified as the sEMG signal of the target muscle.

C. Data Analysis

The purpose of this study was to estimate dynamic changes in muscle activities as pitch increases during phonation. First, the sEMG signal of the target muscle and corresponding speech signal were segmented into a series of overlapping short-time waveforms by a 50-ms rectangular window with a sliding time step of 15 ms. Then, sEMG features and acoustic features were extracted from each segment and intercompared.

1) sEMG Feature Extraction: For the sEMG signal, two time-domain features (including root mean square rms and mean absolute value MAV) and three frequency-domain features (including median frequency MDF, mean frequency MNF, and frequency ratio FR) were selected and extracted as follows.

In the time domain, rms was defined as the square root of the average power of the sEMG signal amplitude, while MAV referred to the mean of the absolute value of the sEMG signal amplitude [33]. For each segment of the sEMG signal, rms and MAV are calculated by

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \tag{1}$$

$$\text{MAV} = \frac{1}{N}\sum_{i=1}^{N} |x_i| \tag{2}$$

where x_i is the sEMG signal and N is the signal length.

In the frequency domain, MDF was defined as the frequency at which the power spectral density (PSD) of the sEMG signal was divided into two equal parts, while MNF was calculated from the sum of the product of the EMG power spectrum and the frequency divided by the total power spectrum [33], [34]. Taking the MNF as the segment point, FR was determined as the ratio of PSD components between the low-frequency and high-frequency bands to distinguish the contraction and relaxation states [33]. The equations of the frequency-domain features are given by

$$\sum_{j=1}^{\mathrm{MDF}} P_j = \sum_{j=\mathrm{MDF}}^{M} P_j = \frac{1}{2}\sum_{j=1}^{M} P_j \tag{3}$$

$$\mathrm{MNF} = \frac{\sum_{j=1}^{M} f_j P_j}{\sum_{j=1}^{M} P_j} \tag{4}$$

$$\mathrm{FR} = \frac{\sum_{j=1}^{\mathrm{MNF}} P_j}{\sum_{j=\mathrm{MNF}}^{M} P_j} \tag{5}$$

where f_j and P_j are frequency and amplitude of the PSD of sEMG signal, respectively.

In general, rms and MAV can reflect the strength of muscle contraction, which is related to motor unit recruitment and the degree of synchronization of firing rate [35]. MDF, MNF, and FR are related to the conduction velocity of motor unit action potential, the firing rate of activated motor units, and the synchrony of the excitation rhythm [35]. In general, when the level of muscle activity increases, the number of motor units and firing rate increase, which leads to an increase in rms, MAV, MDF, and MNF but a decrease in FR.

2) Acoustic Feature Extraction: To investigate speech control by CT and LCA muscles, two acoustic features, including fundamental frequency ($F0$) and closed quotient (CQ), were extracted from speech signals. $F0$ was directly controlled by CT muscle and also regulated by LCA muscle [6], [13], while CQ was primarily controlled by TA muscle [8]. Although CQ had no explicit and significant correlation with CT and LCA muscles, this can be indirect evidence to determine whether the estimated muscle activity was interfered with TA muscle [8].

In this study, $F0$ was calculated using open-source software, VoiceSauce [36]. For CQ, the glottal flow waveform was first estimated from the speech signal using the linear predictive coding and inverse filter methods [37]. Then, CQ was calculated as the percentage of the glottal cycle in which the glottal flow was within 20% of the peak-to-peak value of the glottal flow waveform [38].

III. Method Validation

A. Validation Experiment

In order to validate the sEMG method proposed in this study, LEMG (as a gold standard) and sEMG signals were simultaneously recorded by the DataLINK for comparative analysis. Due to the invasive nature of the LEMG examination, seven subjects (six males and one female) participated in this validation experiment. During the trial, the subject was placed in the supine position with the neck fully exposed and asked to produce the vowel/i:/in a comfortable manner. The LEMG acquisition of CT or LCA muscle was performed by an experienced laryngologist (second author) using concentric needle electrodes. At the same time, the sEMG signals of the ipsilateral CT and LCA muscles were recorded and estimated, respectively. Finally, 191 sets of valid data were obtained in this experiment.

Because sEMG and LEMG measured target muscle signals with different sensors and at different positions, the time-domain features, which were closely associated with signal amplitude, were not appropriate as a reference for assessing muscle activity level when comparing these two signals. In contrast, the frequency-domain features were less affected by signal propagation. As a result, the MDF was selected as a representative parameter to evaluate the similarity and synchronicity between sEMG and LEMG signals. Furthermore, the MDF waveforms of sEMG and LEMG were processed with min-max normalization and matched using derivative dynamic time wrapping (DDTW) [39]. The matching degree was quantified by calculating the wrap path distance, with a smaller distance indicating a better match.

B. sEMG and LEMG Comparison

Because LEMG captures electrical signals in a small region near the target muscle and is less affected by surrounding muscles, it can be used as a gold standard for accurate evaluation of extracted muscle activity. Fig. 3 shows a typical comparison of time-domain waveforms and frequency-domain features between sEMG and LEMG signals from one subject. First, although it was difficult to evaluate the similarity of the muscle activities from the time-domain waveforms, the sEMG and LEMG signals had good temporal synchronization. Second, normalized MDF waveforms illustrated the similarity and difference of the sEMG and LEMG signals. For the same muscle [i.e., CT in Fig. 3(a) and LCA in Fig. 3(d)], the normalized MDF waveforms of the sEMG and LEMG signals had a high degree of similarity in the curve trend with a small wrap path distance (i.e., 0.0039 for CT and 0.0053 for LCA), which indicated a high level of pattern matching between muscle activities extracted from the sEMG and LEMG signals. In contrast, for the different muscles [i.e., Fig. 3(b) and (c)], the sEMG signals had observable differences in the shape of the normalized MDF curve with the LEMG signals. Also, the wrap path distances between the different muscles (i.e., 0.0086 and 0.0213) were much higher than those in the same muscle conditions, suggesting lower pattern matching between the LEMG and sEMG in different muscle conditions.

Furthermore, Fig. 4 shows the statistical results of wrap path distances between the sEMG and LEMG signals across all subjects. First, taking the LEMG signal as a reference (i.e., CT or LCA muscle), the sEMG signals of different muscles had significant statistical differences in the wrap path distances with the LEMG signals. This outcome demonstrated that the sEMG signals in CT mode and LCA mode captured distinct activities from different muscles. Second, within the same muscle, the sEMG signals displayed a relatively small wrap path distance (averaged <0.005) from the LEMG signals, which implied that the sEMG signals were capable of capturing the similar features of muscle activity as the LEMG signals. Therefore, the muscle activity estimated using the sEMG method proposed in this study was consistent with that using conventional LEMG signals. This demonstrates the

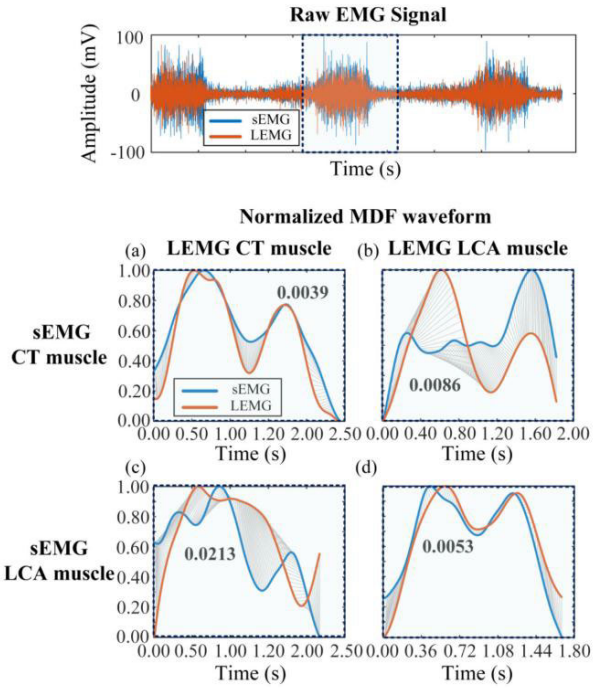

Fig. 3. Examples of temporal waveforms (top panel) and normalized MDF curves (bottom panel) of sEMG and LEMG signals. In the bottom panel, normalized MDF curves of sEMG signals for CT muscle are compared, respectively, with LEMG signals of (a) CT muscle and (b) LCA muscle, and normalized MDF curves of sEMG signals for LCA muscle are also compared with LEMG signals of (c) CT muscle and (d) LCA muscle number in each subfigure of the bottom panel refers to the wrap path distance between the two normalized MDF curves.

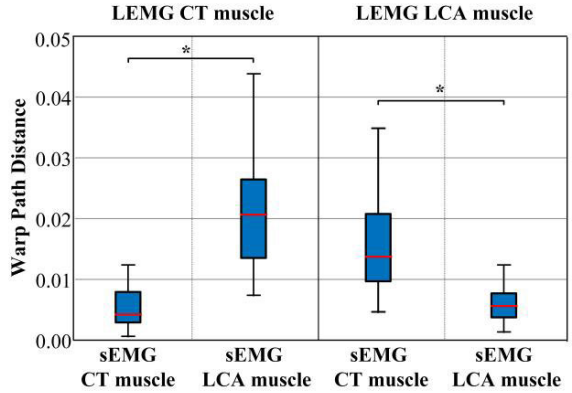

Fig. 4. Wrap path distance between the normalized MDF waveforms of sEMG and LEMG signals. The * indicates a significant difference (p <0.05) according to the Kruskal–Wallis test.

feasibility of using the sEMG method for noninvasive and quantitative investigation of the ILM activities during speech control.

C. Reliability Assessment

To evaluate the stability of the sEMG measurement and processing method, the time-domain feature MAV and frequency-domain feature MDF were selected to perform a consistency analysis. Specifically, the intraclass correlation coefficient (ICC) was calculated on repeated measurements from the same subject in all phonation experiments (see Section II-A3).

As clearly listed in Table II, all the ICCs were higher than 0.6, and 14 of 16 subjects (87.5%) had ICC values higher than 0.8, which shows excellent agreement among multiple repeated measurements. This result not only validates the high reliability of the proposed sEMG method, but also provides support to the credibility of the research findings.

TABLE I
TECHNICAL PARAMETERS OF THE sEMG SENSORS

Frequency range of interest	15-450 Hz
Input impedance	10^{15} ohms
Power supply	0 to 4,950 mV DC
Power supply tolerance	± 2%.
Current supply	< 20 mA
Maximum permissible input voltage	20 Vdc, 8 Vac
Digital resolution	13-bit ADC giving +/- 4000 counts
Sampling rate	1000 Hz

TABLE II
INTRACLASS CORRELATION COEFFICIENTS OF REPEATED MEASUREMENTS FROM EACH SUBJECT

Subject ID	ICC		Subject ID	ICC	
	MAV	MDF		MAV	MDF
Sub1(M)	0.823	0.865	Sub9(M)	0.862	0.829
Sub2(M)	0.809	0.749	Sub10(M)	0.941	0.856
Sub3(M)	0.917	0.773	Sub11(M)	0.967	0.848
Sub4(F)	0.627	0.907	Sub12(F)	0.966	0.970
Sub5(F)	0.832	0.844	Sub13(M)	0.944	0.852
Sub6(M)	0.773	0.908	Sub14(M)	0.868	0.946
Sub7(F)	0.920	0.964	Sub15(F)	0.909	0.877
Sub8(F)	0.842	0.946	Sub16(F)	0.944	0.871

All results demonstrated statistical significance ($p < 0.001$).

IV. RESULTS

A. Relationships Between sEMG Features and F0

Based on data from Exp 1 and Exp 2, the relationships between sEMG features and acoustic features for CT muscle and LCA muscle were obtained, respectively. Fig. 5 shows the dynamic changes of each sEMG feature with increasing $F0$ for different subjects and muscles. In most cases, sEMG features had a significant correlation with $F0$.

For CT muscle (i.e., Exp 1), consistent relationships between frequency-domain features and $F0$ were observed in both male and female subjects. As the $F0$ increased, MDF and MNF increased, but FR decreased in the overall trend. This result indicated that increased $F0$ was regulated by the increase of CT muscle activity, which was in agreement with the experimental observations in previous studies [13], [28]. However, time-domain features showed uncertain variability with $F0$ changes in different subjects; for example, rms and MAV had positive correlations with $F0$ in the male subject but negative and insignificant correlations in the female subject. These negative or insignificant outcomes showed a decrease or less change in the strength of CT muscle contraction, which

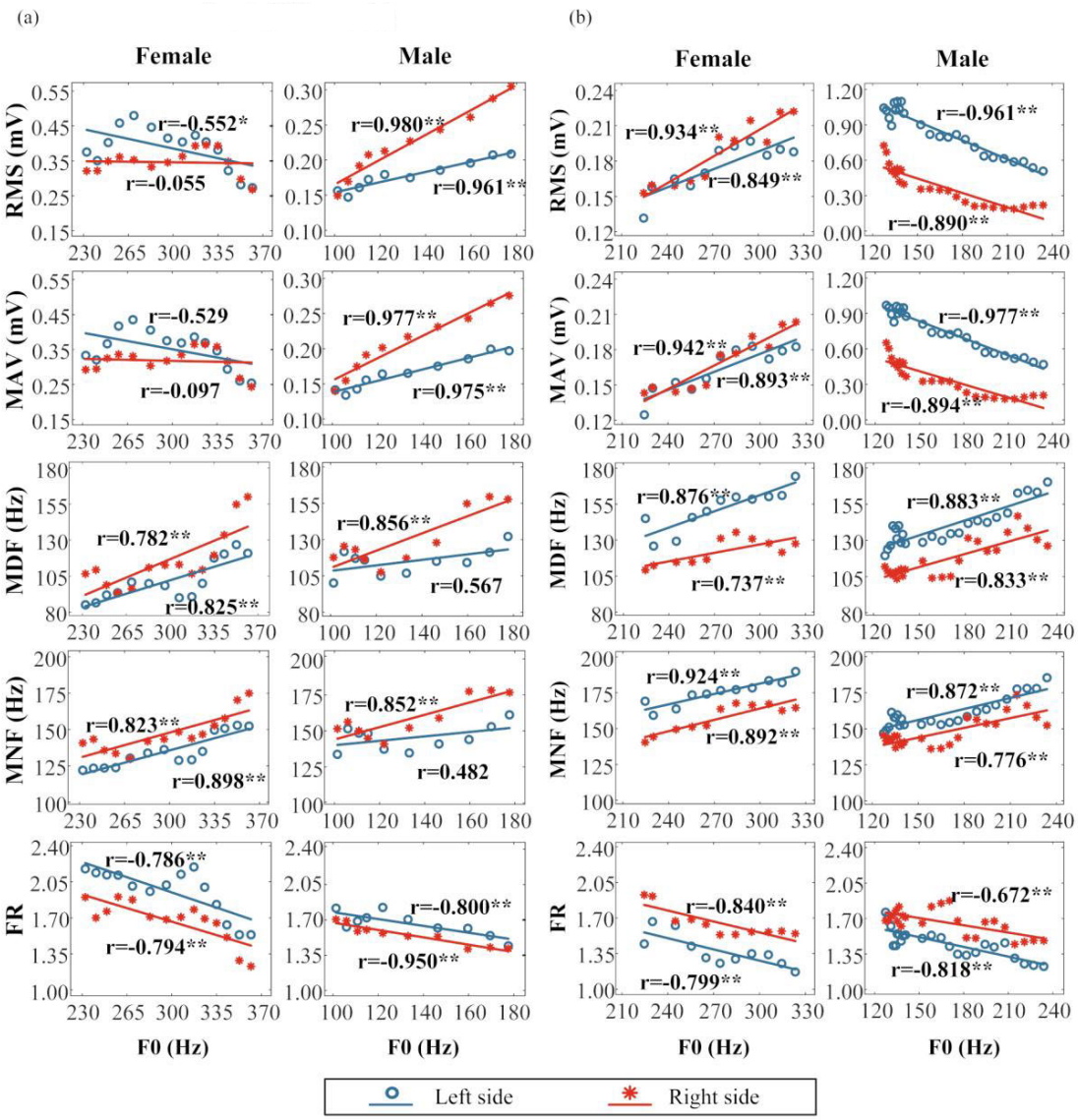

Fig. 5. Relationships of sEMG features with $F0$ for different subjects in (a) Exp 1 and (b) Exp 2. In each subfigure, the scatter plots show extracted sEMG features on the left (blue color) and right (red color) sides of the larynx, while the solid lines represent linear fitting curves of the scatter data. Pearson's correlation coefficients between sEMG features and $F0$ are shown, with statistical significance indicated (* denotes $p < 0.05$, ** denotes $p < 0.01$), and p-values were computed using a two-tailed t-test based on the Pearson correlation coefficient.

was not expected according to the effects of CT muscle in the pitch control [13]. In these cases, an obvious upward movement of the larynx was observed, which may increase the distance between the sEMG sensor and the target muscle, thus leading to higher attenuation in the propagation and a lower amplitude of the sEMG signal. When the amplitude attenuation caused by upward movement is not less than the amplitude increase caused by increased muscle activation, the time-domain feature is finally negatively or not correlated with the $F0$.

In Exp 2, Fig. 5 illustrates the similar relationships between frequency-domain features of LCA muscle and $F0$ as in Exp 1. For all subjects, MDF and MNF were positively related to $F0$, while FR had a negative correlation with $F0$. This result indicated LCA muscle activation gradually went up to increase voice pitch, which was also consistent with the effect of LCA muscle in the pitch control [6], [13]. Likewise, time-domain features (i.e., rms and MAV) also showed different correlations with $F0$ in different subjects, but the difference was independent of gender, which should be relevant to larynx movement during pitch control.

Since the sensors were placed in the same mode on both sides of the larynx in Exp 1 and Exp 2, Fig. 5 also compares the features of bilateral sEMG signals. For both the CT and LCA muscles, despite differences in the absolute values of each sEMG feature between the left and right sides, the dynamic changes in sEMG features with increasing $F0$ were similar in magnitude and synchronized in time on both sides. Especially for frequency-domain features, the fitting curves of the two sides were parallel, indicating a consistent pattern in pitch control for bilateral muscles. In contrast, larger differences occurred in the time-domain features between the left and right sides, which might be due to the asymmetry in anatomical structure, tissue properties, sensor placement, and larynx movement during phonation. These physiological factors have important impacts on the propagation of the

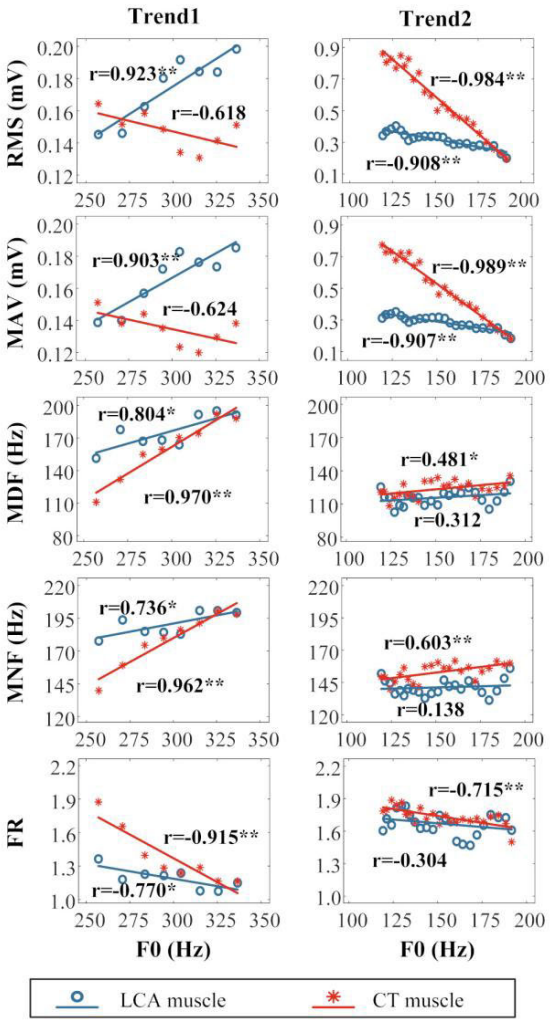

Fig. 6. Relationships of sEMG features with $F0$ for different muscles in Exp 3. In each subfigure, the scatter plots show extracted sEMG features of LCA (blue color) and CT (red color) muscles, while the solid lines represent linear fitting curves of the scatter data. Pearson's correlation coefficients between sEMG features and $F0$ are shown, with statistical significance indicated (* denotes $p < 0.05$, ** denotes $p < 0.01$), and p-values were computed using a two-tailed t-test based on the Pearson correlation coefficient.

electrical signals. The differences in these factors on the left and right sides will cause different amplitudes of the sEMG signals, thus generating different time-domain features.

B. Different Patterns Between CT and LCA Muscles

Based on the results of bilateral symmetry in Exp 1 and Exp 2, it was assumed in Exp 3 that the activity patterns of each muscle on the left and right sides were consistent. Fig. 6 shows two typical comparisons of changes in sEMG features with increasing $F0$ between CT and LCA muscles. First, similar to Exp 1 and 2, rms and MAV showed varied patterns across subjects for CT and LCA muscles (i.e., opposite changes in trend 1 and same changes in trend 2). In contrast, MDF and MNF had consistent positive relationships with $F0$, indicating that both CT and LCA muscles increased activities as pitch increased. Second, CT and LCA muscles had observable differences in the increasing patterns of frequency-domain features. For example, MDF and MNF in trend 1 had larger slopes of the linear fitting curves for CT muscle than LCA muscle. Even if the fitting slopes were comparable, like in trend 2, the changes in the frequency-domain features (see scatter plots) for CT and LCA muscles were not in phase, which could also be observed in trend 1.

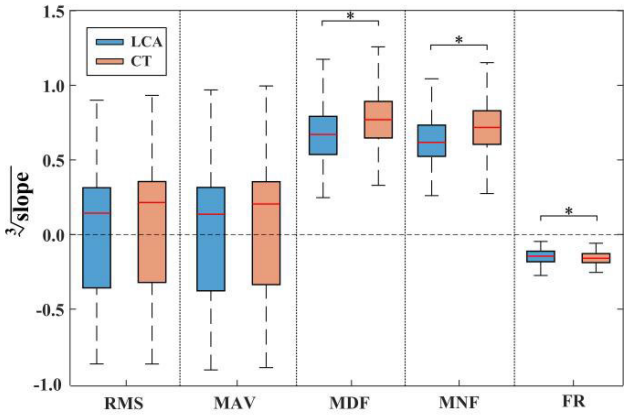

Fig. 7. Comparison of the slopes of linear fitting curves between LCA and CT muscles in Exp 3. The vertical axis shows the cubic root of the slope, and the * indicates a significant difference ($p < 0.05$) according to the Mann–Whitney U test.

For further investigation of different patterns between CT and LCA muscles, Fig. 7 illustrates the slopes of the fitting curves for each feature. Because rms, MAV, and FR were numerically much smaller than MDF and MNF, the cubic root of the fitting slopes was shown on the vertical axis to make all the data in a comparable range. Although there was no statistical difference in the slopes of the time-domain features between CT and LCA muscles, the frequency-domain features showed significantly different fitting slopes between the two muscles. Moreover, the fitting slopes of MDF and MNF in CT muscle were remarkably larger than those in LCA muscle. This finding suggested that the CT muscle experienced a greater increase in the activation degree and exerted a stronger influence in $F0$ control. This result was consistent with previous research on the contributions of the CT and LCA muscles to pitch control [40].

C. Relationships Between sEMG Features and CQ

Fig. 8 shows typical relationships between sEMG features and CQ for CT and LCA muscles from all subjects. In most cases, the linear fitting curves were flatter, with absolute values of correlation coefficients between sEMG features and CQ less than 0.4, meaning that sEMG features of CT and LCA muscles had no significant correlations with CQ. In the other cases, although the correlations between sEMG features and CQ were significant, the distributions of the scatter points were dispersed. Meanwhile, for example, the fitting slope of the rms curve for CT and LCA muscle was positive in trend 2 but negative in trend 3, which suggested uncertain variability of sEMG features with CQ changes. Consequently, the results demonstrated CT and LCA muscle activation had no significant correlation with CQ control, which was in agreement with the findings in previous studies [8].

V. Discussion

In this work, a noninvasive method using two-channel sEMG signals was designed to extract the dynamic activities of ILMs (i.e., CT and LCA) during speech production.

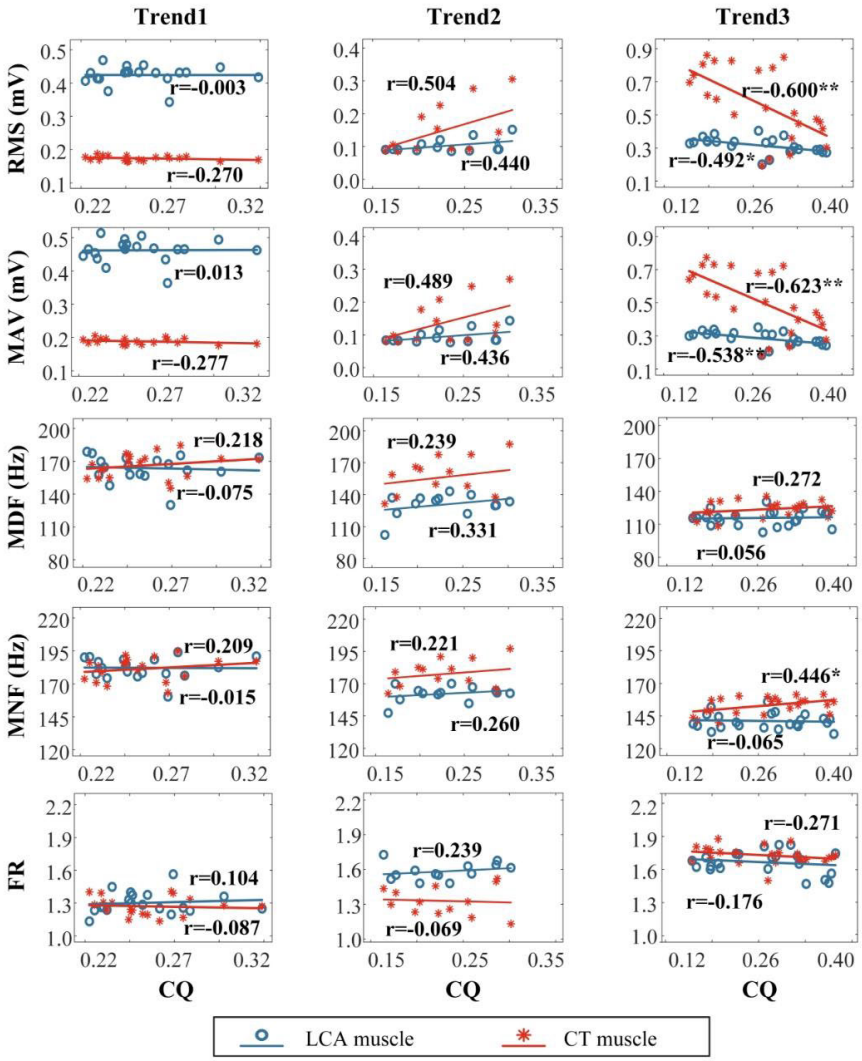

Fig. 8. Relationships of sEMG features with CQ in Exp 3. In each subfigure, the scatter plots show extracted sEMG features of LCA (blue color) and CT (red color) muscles, while the solid lines represent linear fitting curves of the scatter data. Pearson's correlation coefficients between sEMG features and CQ are shown, with statistical significance indicated (* denotes $p < 0.05$, ** denotes $p < 0.01$), and p-values were computed using a two-tailed t-test based on the Pearson correlation coefficient.

Due to the small size of the larynx and the close distance between different laryngeal muscles, a specific placement of the main differential sensor was applied to maximize the sEMG signals of the target muscle. Moreover, the interference signals (mainly from SH and ST muscles) were captured using the secondary sensor and removed through the Fast ICA algorithm. The goal of this study was to evaluate the feasibility and effectiveness of sEMG signal acquisition and processing method in the noninvasive and quantitative measurement of laryngeal muscle activity.

Regarding sEMG signal acquisition, considering the fact that the distance between CT and LCA muscles was comparable to the size of the differential sensor, the main sensor was placed on the target muscle with its axial direction parallel to the fiber orientation. According to the theory of EMG propagation and measurement using differential electrodes, the electrical signal from the muscle that is not parallel to the sensor placement is largely attenuated. As a result, this placement mode was able to reduce the crosstalk between the sEMG signals of CT and LCA muscles, which was demonstrated by the significant differences in features of the sEMG signals captured by the two modes (see Figs. 3 and 6). Also, the sEMG signals of the secondary sensor were different in waveforms and features from the signals of the main sensor (see Fig. 2), indicating the necessity of the secondary sensor in the removal of noise and interference components. In addition, although the sensor placement was performed by a skilled technician who was familiar with the anatomical structure of the larynx, it is not guaranteed that the position of the sensor is accurate. However, the results showed that the sEMG signals had a good agreement with the LEMG signals and a good consistency in control pattern across different subjects. Furthermore, the ICCs (see Table II) demonstrated excellent consistency and reproducibility during repeated measurements on the same subject. These outcomes indicate that minor position deviations caused by manual placement are tolerable and do not exert a significant impact on the accuracy of the measurements. This further validates the robustness and reliability of the proposed sEMG method.

For the target muscle, the validity of the extracted sEMG signals was verified directly by comparing sEMG signals with LEMG signals, which is a gold standard for clinical

examination of laryngeal muscles. Due to differences in measurement methods and positions, the sEMG signal was not comparable with the LEMG signal in the time domain. However, the normalized MDF waveforms and wrap path distances showed good synchronization and a strong match in curve changes between the sEMG signal and LEMG signals (see Figs. 3 and 4). This result confirms that the sEMG signals measured using the method in this work are consistent with the LEMG signals in the frequency-domain features.

Meanwhile, the validity of the extracted sEMG signals can also be indirectly verified by comparing the sEMG features and acoustic features. According to current knowledge, $F0$ was generally controlled by CT, LCA, and TA muscles, while CQ was mainly regulated by TA muscle activity [8], [13]. In this work, the extracted sEMG signals in both CT mode and LCA mode showed significant and consistent patterns between the frequency-domain features and $F0$ (see Fig. 5), but had no significant correlation between the sEMG features and CQ (see Fig. 8). Also, both CT and LCA muscles had increasing activity with increasing $F0$, which was in line with the findings in previous studies [13]. Consequently, it is demonstrated that the extracted sEMG signals are highly related to the target muscle activities and are not interfered with by TA muscle. Furthermore, in a condition with increasing $F0$, the activities of SH and ST muscles normally decrease first and then increase [40]. This pattern was not observed in the sEMG features of the target muscles, indicating an effective removal of the interference signals generated by external laryngeal muscles.

Although both CT and LCA modes showed a similar upward trend in sEMG features (i.e., frequency-domain features) with increasing $F0$ (see Figs. 5 and 6), the relationship between sEMG features and $F0$ differed significantly between the CT and LCA muscles (see Fig. 6). Specifically, the absolute value of the linear slope of the positive correlation between the frequency-domain features and $F0$ was statistically higher in the CT mode than in the LCA mode (see Fig. 7), which indicates that CT muscle had a greater overall increase of muscle activity than LCA muscle under the same $F0$ increase condition. This result is consistent with the fact that CT muscle plays a more important role than LCA muscle in pitch control [41], which indirectly validates that the extracted sEMG signals in the CT and LCA modes were highly related to the muscle activities of CT and LCA muscles, respectively. It is also demonstrated that the specific placement mode and extraction method are able to effectively separate the sEMG signals of CT and LCA muscles, allowing for accurate estimation of their muscle activities during phonation.

In the study, two time-domain features and three frequency-domain features of sEMG signals were calculated and compared with the acoustic features. It was found that frequency-domain features (i.e., MDF, MNF, and FR) showed a consistent variation pattern with increasing $F0$ across subjects, while time-domain features (i.e., rms and MAV) had an uncertain correlation with the change in $F0$ (see Figs. 5 and 6). Since the amplitude of sEMG signals is closely associated with signal transmission, the results suggest that the changes in the relative position between the target muscle and the sensor during phonation have a significant impact on the time-domain features. In contrast, the frequency-domain features exhibit greater stability during phonation and are less influenced by the amplitude variations of the sEMG signal caused by larynx movement. However, the frequency-domain features were also influenced by external factors. For instance, changes in electrode position and angle, high skin impedance, and subtle motion artifacts may lead to attenuation or distortion in certain frequency bands, thereby affecting the stability of sEMG signal frequency-domain features. Nevertheless, frequency-domain features demonstrate adequate stability during phonation in this study and exhibit strong consistency between the sEMG and LEMG signals (see Figs. 3 and 4). As a result, from the point of view of practical application, the frequency-domain features are more appropriate and reliable than the time-domain features as a parameter to evaluate the activity of laryngeal muscles. Nonetheless, it should be noted that the time-domain features have an advantage in capturing transient signals, which is fundamental in evaluating rapid muscle control during speech production.

Under the condition of pitch control in chest voice, this work verified the effectiveness of the proposed sEMG method for noninvasive measurement of LCA and CT muscle activities during phonation. However, study with a small sample size were carried out because of difficulties in subject recruitment due to the invasive nature of LEMG. Although the dataset size was sufficient to show statistical significance and consistency in the experimental results, the sEMG method still needs to be validated in a larger population in future work to fully assess its potential for clinical applications. Meanwhile, this study was only in the view of the normal subjects; pathological conditions (such as CT or LCA paralysis) need further experimental data to confirm the feasibility of this method in clinical applications.

VI. Conclusion

In this study, a two-channel sEMG method was developed to extract the sEMG signals of specific ILMs (i.e., CT muscle and LCA muscle) during in vivo human phonation. The designed sensor placement mode and processing approach could maximize the sEMG signal of the target muscle and eliminate interfering signals from the surrounding muscles. The extracted sEMG signals captured the similar features of muscle activity as the LEMG signals. Especially, the frequency-domain features of sEMG signals provided a quantitative evaluation of the dynamic activity of CT and LCA muscle muscles during speech control, demonstrating a stable and consistent positive correlation with voice pitch. The CT muscle exhibited greater activity and played a more significant role in pitch control, which was consistent with the findings in previous experimental studies using animal or excised models. Therefore, this work confirms that the sEMG method is a feasible and reliable tool for noninvasive measurement and quantitative evaluation of ILMs activity during speech production, thus providing a promising technology to investigate the neuromuscular mechanism of voice control and diagnose laryngeal disorders.

Acknowledgment

The authors had full access to all of the data, devices, and materials used in this study and take complete responsibility for the integrity of the data and the accuracy of the data analysis and interpretation of outcomes.

REFERENCES

[1] Z. Zhang, "Mechanics of human voice production and control," *J. Acoust. Soc. Amer.*, vol. 140, no. 4, pp. 2614–2635, 2016.

[2] P. Pillutla, N. K. Reddy, P. Schlegel, Z. Zhang, and D. K. Chhetri, "Control of pre-phonatory glottal shape by intrinsic laryngeal muscles," *Laryngoscope*, vol. 133, no. 7, pp. 1690–1697, Sep. 2022.

[3] S. Hewavitharanage, J. Gubbi, D. Thyagarajan, K. Lau, and M. Palaniswami, "Estimation of vocal fold plane in 3D CT images for diagnosis of vocal fold abnormalities," in *Proc. 37th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Milan, Italy, Aug. 2015, pp. 3105–3108.

[4] C. E. Stepp, R. E. Hillman, and J. T. Heaton, "Use of neck strap muscle intermuscular coherence as an indicator of vocal hyperfunction," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 18, no. 3, pp. 329–335, Jun. 2010.

[5] D. K. Chhetri, J. Neubauer, and D. A. Berry, "Neuromuscular control of fundamental frequency and glottal posture at phonation onset," *J. Acoust. Soc. Amer.*, vol. 131, no. 2, pp. 1401–1412, Feb. 2012.

[6] D. K. Chhetri and J. Neubauer, "Differential roles for the thyroarytenoid and lateral cricoarytenoid muscles in phonation," *Laryngoscope*, vol. 125, no. 12, pp. 2772–2777, Jul. 2015.

[7] D. K. Chhetri, J. Neubauer, E. Sofer, and D. A. Berry, "Influence and interactions of laryngeal adductors and cricothyroid muscles on fundamental frequency and glottal posture control," *J. Acoust. Soc. Amer.*, vol. 135, no. 4, pp. 2052–2064, Apr. 2014.

[8] G. Luegmair, D. K. Chhetri, and Z. Zhang, "The role of thyroarytenoid muscles in regulating glottal closure in an in vivo canine larynx model," in *Proc. Meetings Acoust.*, vol. 22, no. 1, Indianapolis, India, 2014, Paper 060007.

[9] R. Islam, M. Tarique, and E. Abdel-Raheem, "A survey on signal processing based pathological voice detection techniques," *IEEE Access*, vol. 8, pp. 66749–66776, 2020.

[10] H. Liu, R. Behroozmand, M. Bove, and C. R. Larson, "Laryngeal electromyographic responses to perturbations in voice pitch auditory feedback," *J. Acoust. Soc. Amer.*, vol. 129, no. 6, pp. 3946–3954, Jun. 2011.

[11] P.-Å. Lindestad, B. Fritzell, and A. Persson, "Quantitative analysis of laryngeal EMG in normal subjects," *Acta Oto-Laryngolog.*, vol. 111, no. 3, pp. 1146–1152, 1991.

[12] K. A. Kochis-Jennings, E. M. Finnegan, H. T. Hoffman, S. Jaiswal, and D. Hull, "Cricothyroid muscle and thyroarytenoid muscle dominance in vocal register control: Preliminary results," *J. Voice*, vol. 28, no. 5, pp. 652.e21–652.e29, Sep. 2014.

[13] V. J. Boucher, C. Ahmarani, and T. Ayad, "Physiologic features of vocal fatigue: Electromyographic spectral-compression in laryngeal muscles," *Laryngoscope*, vol. 116, no. 6, pp. 959–965, Jun. 2006.

[14] L. J. Smith, C. A. Rosen, C. Niyonkuru, and M. C. Munin, "Quantitative electromyography improves prediction in vocal fold paralysis," *Laryngoscope*, vol. 122, no. 4, pp. 854–859, Feb. 2012.

[15] Y. D. Heman-Ackah, S. Mandel, R. Manon-Espaillat, M. Abaza, and R. T. Sataloff, "Laryngeal electromyography," *Otolaryngol. Clinics North Amer.*, vol. 40, no. 5, pp. 1003–1023, Sep. 2007.

[16] G. F. Volk et al., "Laryngeal electromyography: A proposal for guidelines of the European laryngological society," *Eur. Arch. Oto-Rhino-Laryngol.*, vol. 269, no. 10, pp. 2227–2245, May 2012.

[17] R. O'Keeffe, S. Y. Shirazi, S. Mehrdad, T. Crosby, A. M. Johnson, and S. F. Atashzar, "Perilaryngeal-cranial functional muscle network differentiates vocal tasks: A multi-channel sEMG approach," *IEEE Trans. Biomed. Eng.*, vol. 69, no. 12, pp. 3678–3688, Dec. 2022.

[18] A. Blitzer et al., "Recommendations of the neurolaryngology study group on laryngeal electromyography," *Otolaryngology–Head Neck Surgery*, vol. 140, no. 6, pp. 782–793, Jun. 2009.

[19] N. R. Smith, T. Klongtruagrok, G. N. DeSouza, C. R. Shyu, M. Dietrich, and M. P. Page, "Non-invasive ambulatory monitoring of complex sEMG patterns and its potential application in the detection of vocal dysfunctions," in *Proc. IEEE 16th Int. Conf. e-Health Netw., Appl. Services (Healthcom)*, Natal, Brazil, Oct. 2014, pp. 447–452.

[20] P. M. M. Balata, H. J. D. Silva, L. D. A. Pernambuco, J. H. P. D. Oliveira, and S. R. A. D. Moraes, "Normalization patterns of the surface electromyographic signal in the phonation evaluation," *J. Voice*, vol. 29, no. 1, pp. 129.e1–129.e8, Jan. 2015.

[21] R. O'Keeffe, S. Y. Shirazi, S. Mehrdad, T. Crosby, A. M. Johnson, and S. F. Atashzar, "Perilaryngeal functional muscle network in patients with vocal hyperfunction—A case study," in *Proc. 11th Int. IEEE/EMBS Conf. Neural Eng. (NER)*, Baltimore, MD, USA, Apr. 2023, pp. 1–5.

[22] E. Van Houtte, S. Claeys, E. D'haeseleer, F. Wuyts, and K. Van Lierde, "An examination of surface EMG for the assessment of muscle tension dysphonia," *J. Voice*, vol. 27, no. 2, pp. 177–186, Mar. 2013.

[23] M. Zhu et al., "Automatic speech recognition in different languages using high-density surface electromyography sensors," *IEEE Sensors J.*, vol. 21, no. 13, pp. 14155–14167, Jul. 2021.

[24] Z. Deng, X. Zhang, X. Chen, X. Chen, X. Chen, and E. Yin, "Silent speech recognition based on surface electromyography using a few electrode sites under the guidance from high-density electrode arrays," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.

[25] D. J. Bracken, G. Ornelas, T. P. Coleman, and P. A. Weissbrod, "High-density surface electromyography: A visualization method of laryngeal muscle activity," *Laryngoscope*, vol. 129, no. 10, pp. 2347–2353, Jan. 2019.

[26] D. Farina, L. Mesin, S. Martina, and R. Merletti, "A surface EMG generation model with multilayer cylindrical description of the volume conductor," *IEEE Trans. Biomed. Eng.*, vol. 51, no. 3, pp. 415–426, Mar. 2004.

[27] S. Ma et al., "Analytical modelling of surface EMG signals generated by curvilinear fibers with approximate conductivity tensor," *IEEE Trans. Biomed. Eng.*, vol. 69, no. 3, pp. 1052–1062, Mar. 2022.

[28] J. Martínez, O. Valencia, and M. Zañnartu, "High-density electromyography as a method for estimating laryngeal muscle activity: A preliminary study," *Proc. SPIE*, vol. 12567, pp. 373–379, Mar. 2023.

[29] S. S. Azar and D. K. Chhetri, "Phonation threshold pressure revisited: Effects of intrinsic laryngeal muscle activation," *Laryngoscope*, vol. 132, no. 7, pp. 1427–1432, Nov. 2021.

[30] T.-J. Fang et al., "Quantitative laryngeal electromyography assessment of cricothyroid function in patients with unilateral vocal fold paralysis," *Laryngoscope*, vol. 125, no. 11, pp. 2530–2535, Jun. 2015.

[31] X. Xu et al., "Study on normal laryngeal electromyography of thyroarytenoid muscle, cricothyroid muscle, and posterior cricoarytenoid muscle," *Ann. Otol., Rhinol. Laryngol.*, vol. 127, no. 11, pp. 806–811, Sep. 2018.

[32] C. Y. Sai, N. Mokhtar, H. Arof, P. Cumming, and M. Iwahashi, "Automated classification and removal of EEG artifacts with SVM and wavelet-ICA," *IEEE J. Biomed. Health Informat.*, vol. 22, no. 3, pp. 664–670, May 2018.

[33] A. Phinyomark, P. Phukpattaranont, and C. Limsakul, "Feature reduction and selection for EMG signal classification," *Exp. Syst. Appl.*, vol. 39, no. 8, pp. 7420–7431, Jun. 2012.

[34] B. N. Cahyadi et al., "Analysis of EMG based arm movement sequence using mean and median frequency," in *Proc. 5th Int. Conf. Electr. Eng., Comput. Sci. Informat. (EECSI)*, Malang, Indonesia, Oct. 2018, pp. 440–444.

[35] R. Merletti, A. Rainoldi, and D. Farina, "Surface electromyography for noninvasive characterization of muscle," *Exercise Sport Sci. Rev.*, vol. 29, no. 1, pp. 20–25, Jan. 2001.

[36] Y.-L. Shue, P. Keating, and C. Vicenik, "VOICESAUCE: A program for voice analysis," *J. Acoust. Soc. Amer.*, vol. 126, no. 4, p. 2221, Oct. 2009.

[37] H. Pulakka, "Analysis of human voice production using inverse filtering, high-speed imaging, and electroglottography," M.S. thesis, Dept. Comput. Sci. Eng., Hels. Univ. Technol., Espoo, Finland, 2005.

[38] L. Wu and Z. Zhang, "Effects of implant and vocal fold stiffness on voice production after medialization laryngoplasty in an MRI-based vocal fold model," *J. Biomechanics*, vol. 149, Mar. 2023, Art. no. 111483.

[39] W. Li, R. He, B. Liang, F. Yang, and S. Han, "Similarity measure of time series with different sampling frequencies based on context density consistency and dynamic time warping," *IEEE Signal Process. Lett.*, vol. 30, pp. 1417–1421, 2023.

[40] D. K. Chhetri and S. J. Park, "Interactions of subglottal pressure and neuromuscular activation on fundamental frequency and intensity," *Laryngoscope*, vol. 126, no. 5, pp. 1123–1130, Mar. 2016.

[41] B. Roubeau, C. Chevrie-Muller, and J. L. Saint Guily, "Electromyographic activity of strap and cricothyroid muscles in pitch change," *Acta Oto-Laryngolog.*, vol. 117, no. 3, pp. 459–464, Jan. 1997.